3

Functions

Chapter Learning Outcomes

CLO

3

- Distinguish between Functions and relations.
- Define and use the terms function, domain, codomain, range, image, and pre-image.
- Determine if a function is one-to-one (injection).
- Determine if a function is onto (surjection).
- Determine if a function is bijection.
- Describe the connection between bijective functions and inverses.
- Be able to find the inverse of an invertible function.
- Be able to find the composition between two functions.

What is a Function?

A function is a relation in which every input has exactly one output

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What is a Function?

Example

$$R = \{(0,-5),(1,-4),(2,-3),(3,-2),(4,-1),(5,0)\}$$

Is R a relation ?

Is R a function ?

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Is a Relation a Function?

- Focus on the **x-coordinates**, when given a relation

- If the set of ordered pairs has **different** **x-coordinates**,
  it IS A function

- If the set of ordered pairs has **same** **x-coordinates**,
  it is NOT a function

- **Y-coordinates** have **no bearing** in determining functions

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What is a Function?

Example

$$R = \{(-1,-7),(1,0),(2,-3),(0,-8),(0,5),(-2,-1)\}$$

Is R a relation ?

Is R a function ?

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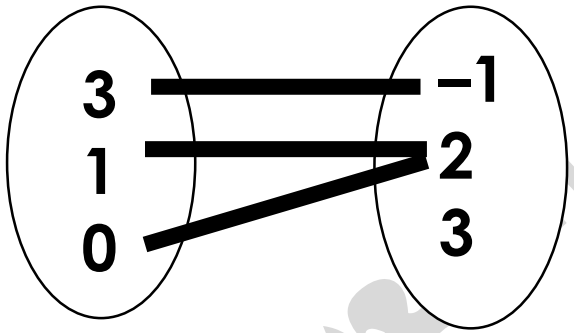
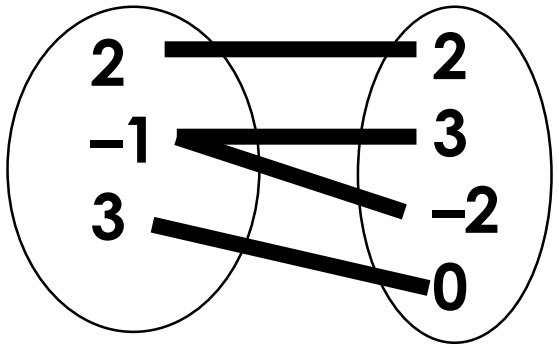
What is a Function?

Example

Which relation mapping represents a function?



R1



R2

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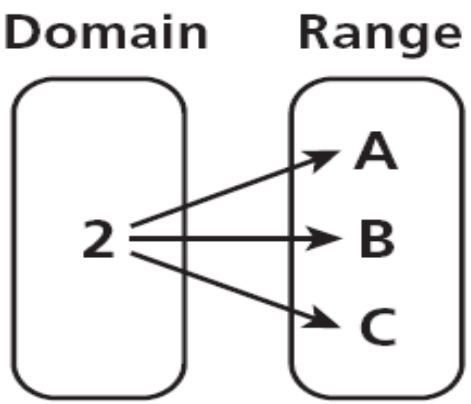
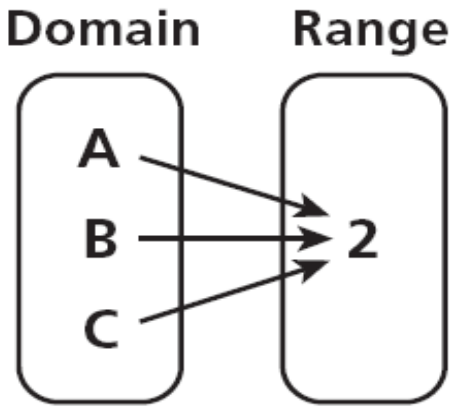
What is a Function?

Example

Which relation mapping represents a function?



A



B

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Vertical Line Test

Vertical Line Test: a relation is a function if a vertical line drawn through its graph, passes through only one point.

"The Pencil Test"

Take a pencil and move it from **left to right (–x to x)**, if it crosses more than one point, it is not a function

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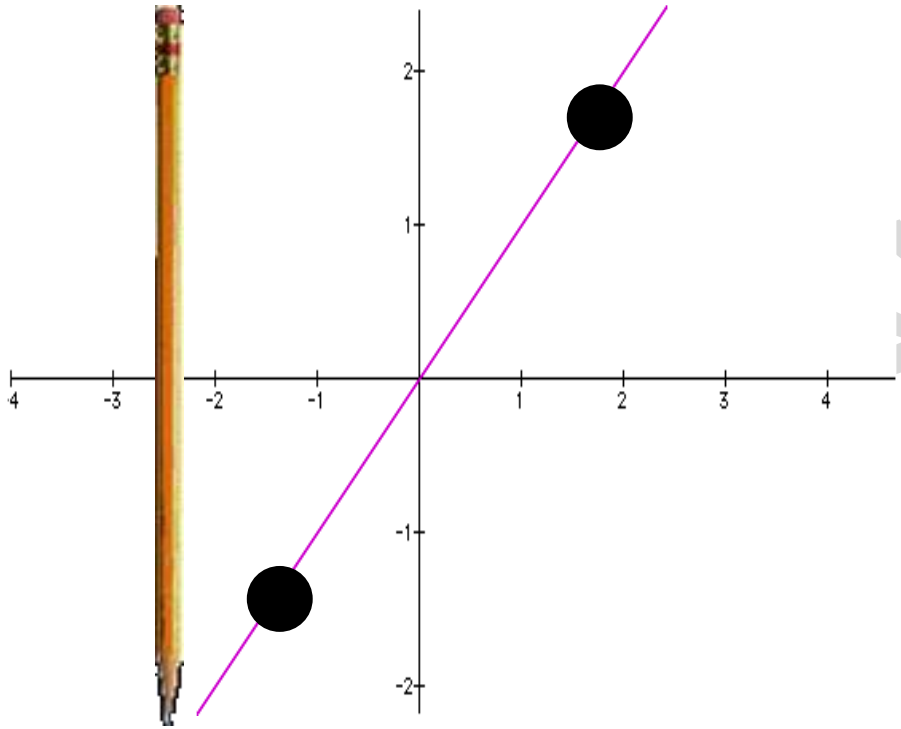
Vertical Line Test

Example

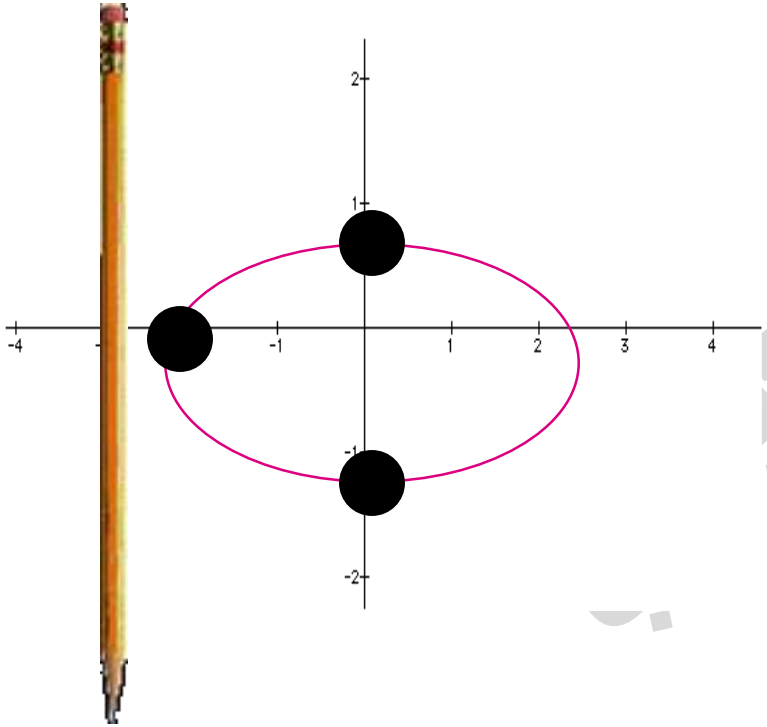
Would this graph be a function?



Vertical Line Test

Example

Would this graph be a function?



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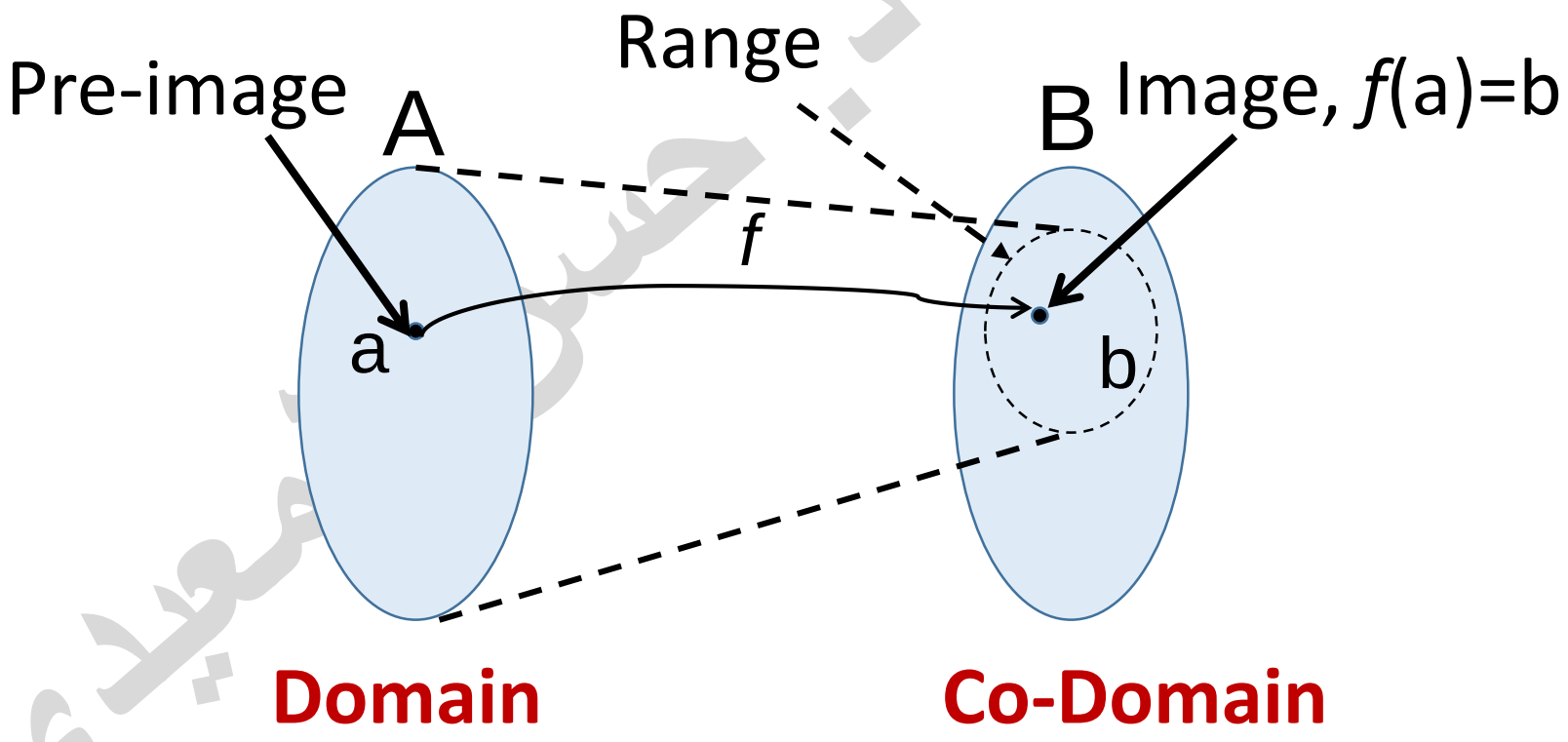
Function: Visualization

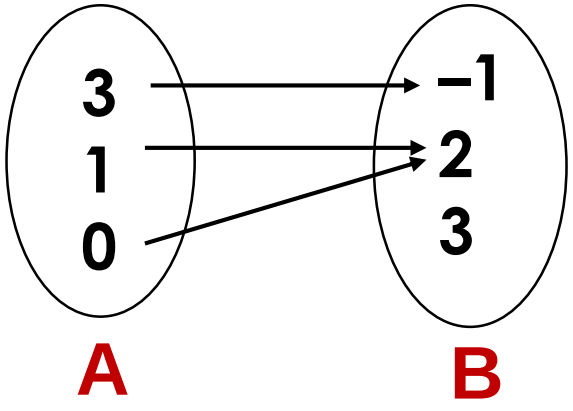


A function, f: A $\rightarrow$ B

Function

EXAMPLE

- Use the following function representation to answer questions below.



$f: A \rightarrow B$

Domain of f

Co-Domain of f

Range of f

Image of 3

Preimage of -1

Preimage of 2

Function Types

Injective (one-to-one)

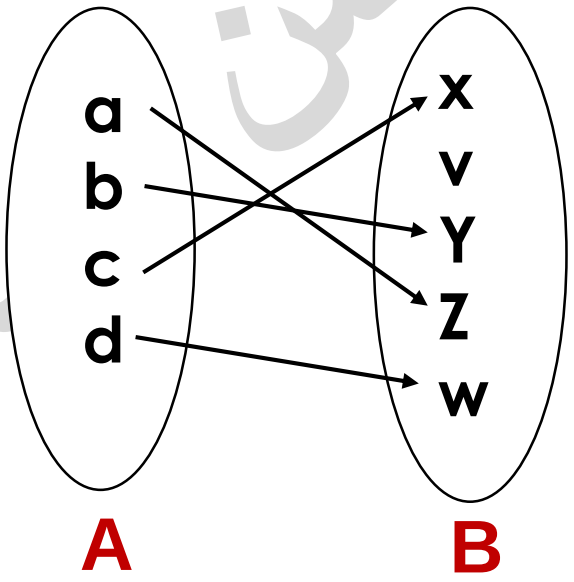
Surjective (onto)

Bijective

Injection

A function f is said to be **one-to-one**, or **injective**, if and only if $f(a) = f(b) \rightarrow a = b$ for all a and b in the domain of f.

EXAMPLE



f: A → B

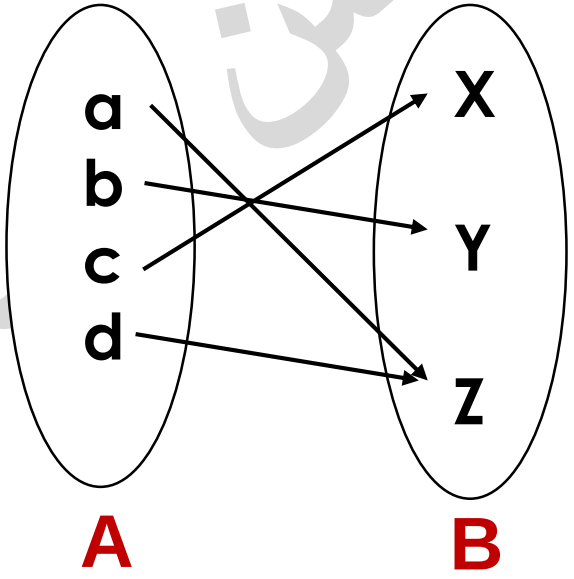
f is an injective function

$$|A| \leq |B|$$

Surjection

A function f is said to be **onto**, or **surjective**, if and only if for every element $b \in B$, there is an element $a \in A$ with $f(a)=b$.

EXAMPLE



$f: A \rightarrow B$

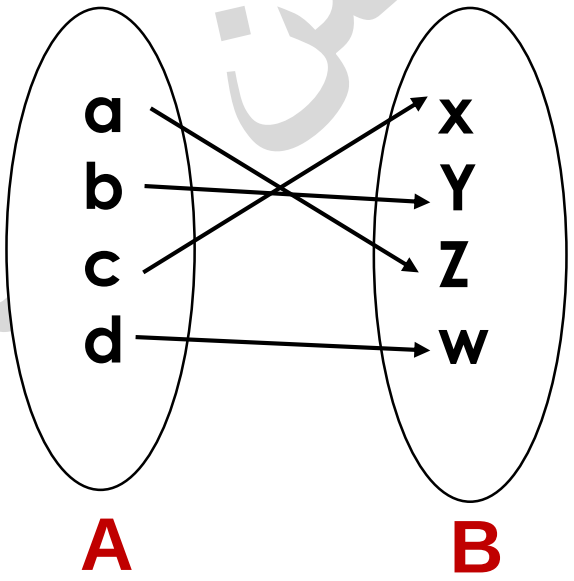
f is a surjective function

$$|A| \geq |B|$$

Bijection

A function f is said to be **bijective**, if it is injective and surjective at the same time.

EXAMPLE



f: A → B

f is a bijective function

$$|A| = |B|$$

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Function Types

Injective (one-to-one)

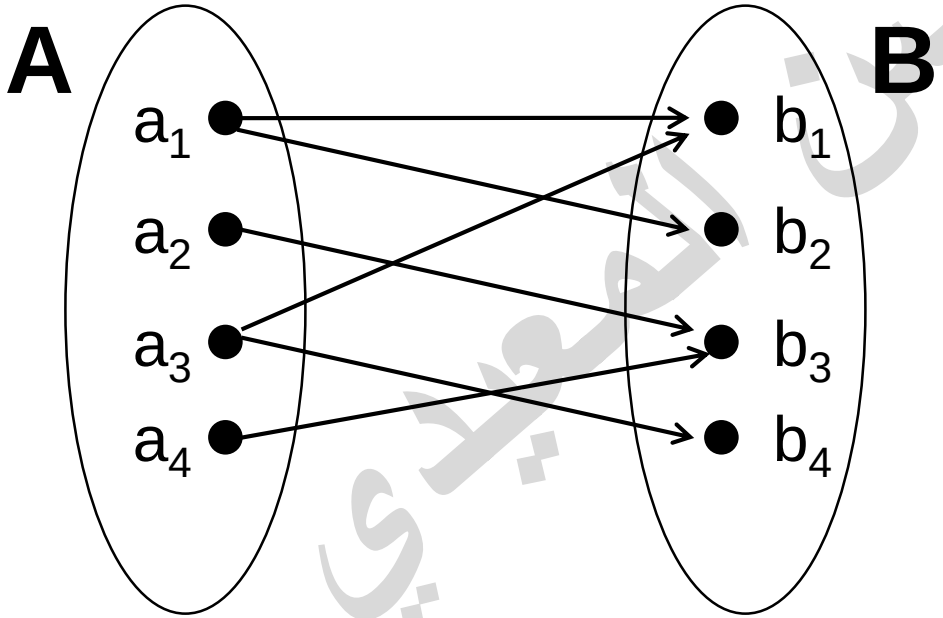
Surjective (onto)

Bijective

Example

EXAMPLE

What is the type of the following function?



Example

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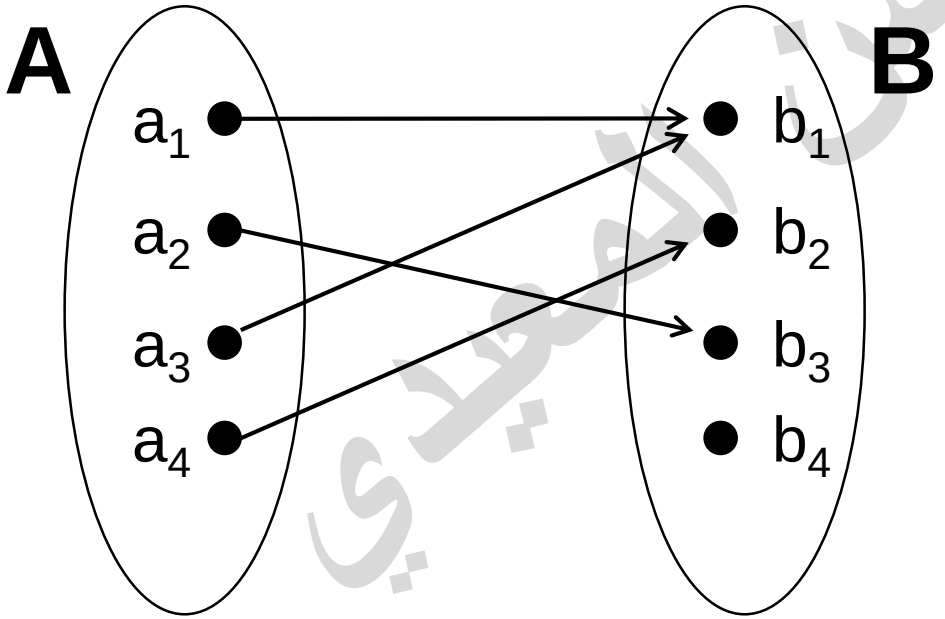

Example

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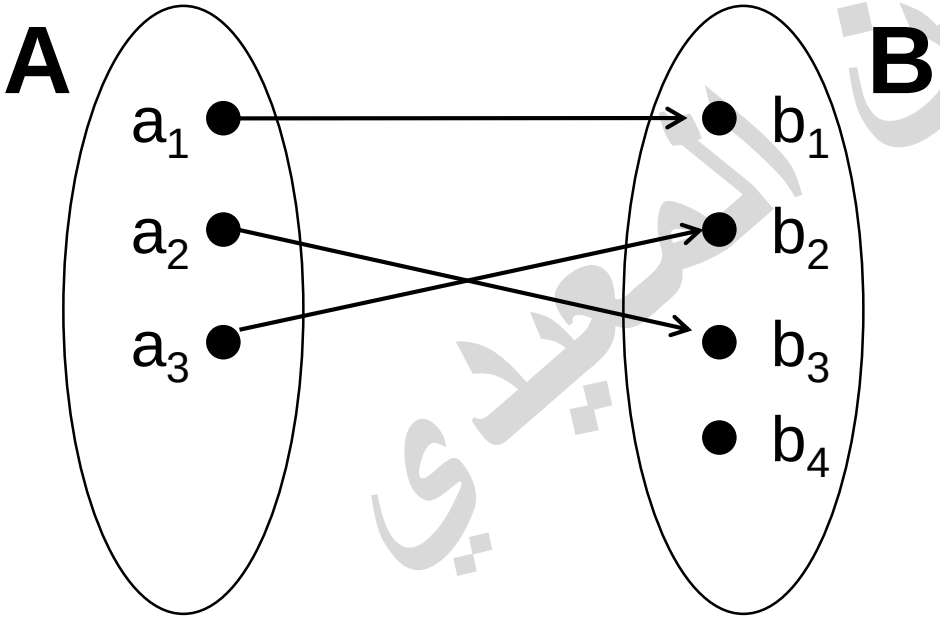

Example

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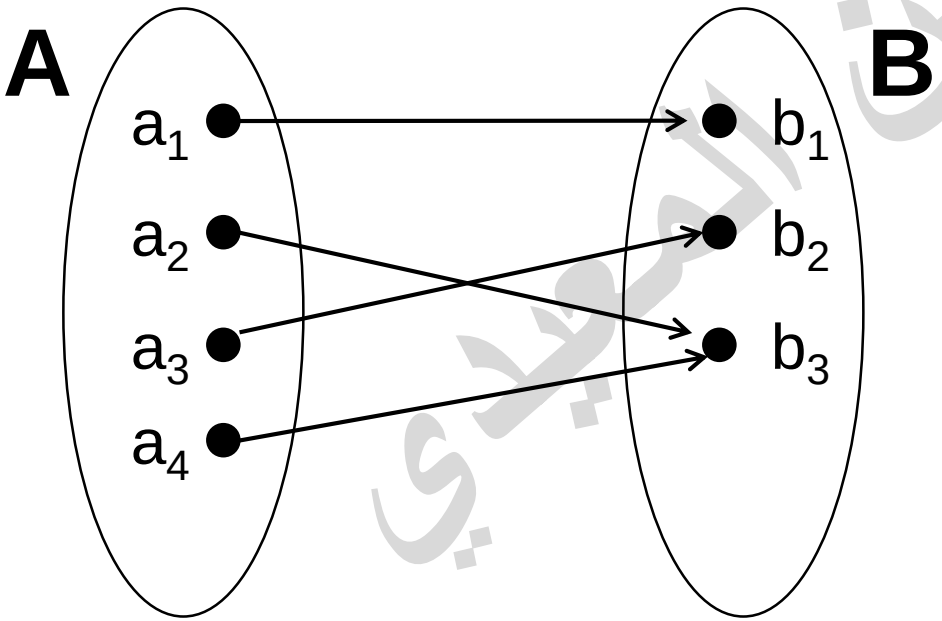

Example

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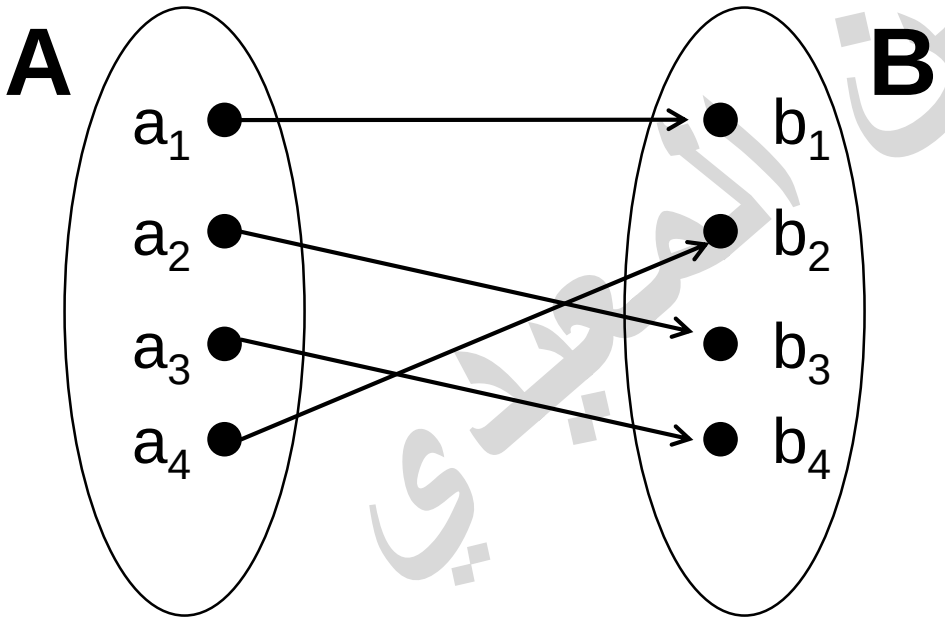


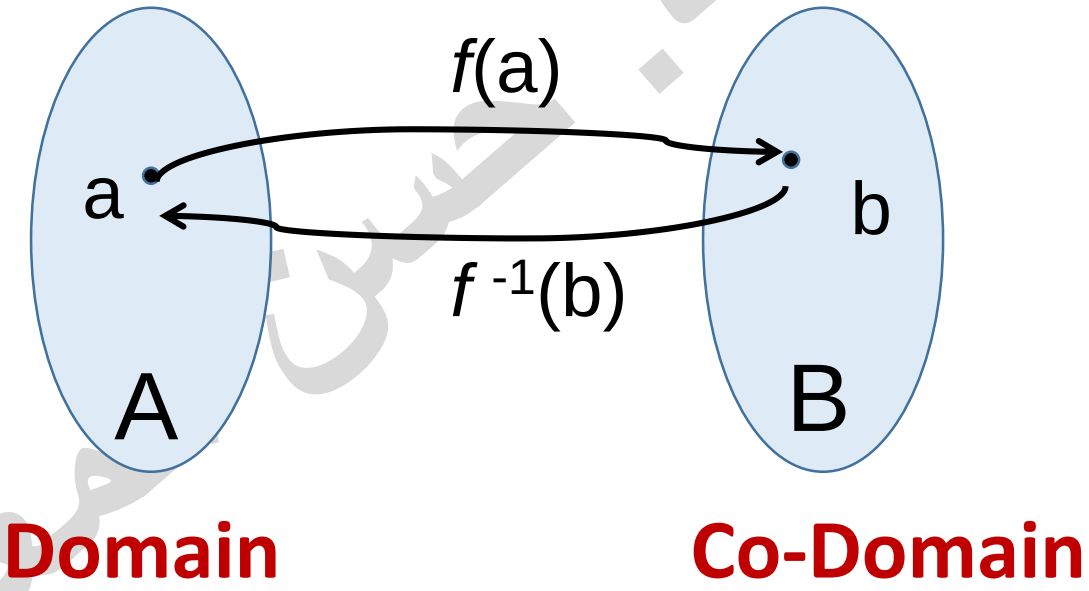
Inverse Function

Let f: A$\rightarrow$B be a **bijection**. The **inverse** function of f is the function that assigns to an element b$\in$B the unique element a$\in$A such that f(a)=b.

- The inverse function is denote f^{-1}.
- When f is a bijection, its inverse exists and

$$f(a)=b \Leftrightarrow f^{-1}(b)=a$$

Inverse Function



A function and its inverse

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Inverse Function

EXAMPLE

Let $f:R \rightarrow R$ be defined by $f(x) = 2x - 3$

What is f^{-1}?

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Inverse Function

EXAMPLE

Let $f:R \rightarrow R$ be defined by $f(x) = x^2$

What is f^{-1}?

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Function Composition

Let g:A→B and f:B →C. The <u>composition</u> of the functions f and g is

$$(f \circ g)(x) = f(g(x))$$

***f*°*g* is read as:**

- *f* circle *g*
- *f* composed with *g*
- *f* following *g*
- *f* of *g*

Function Composition

EXAMPLE

Let f, g be two functions on $R \rightarrow R$ defined by:

$$f(x) = 2x - 3$$

$$g(x) = x^2 + 1$$

What are $f \circ g$?

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Function Composition

EXAMPLE

Let f, g be two functions on $R \rightarrow R$ defined by:

$$f(x) = 2x - 3$$

$$g(x) = x^2 + 1$$

What are $g \circ f$?

Check - Chapter Learning Outcomes

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End of Chapter

3